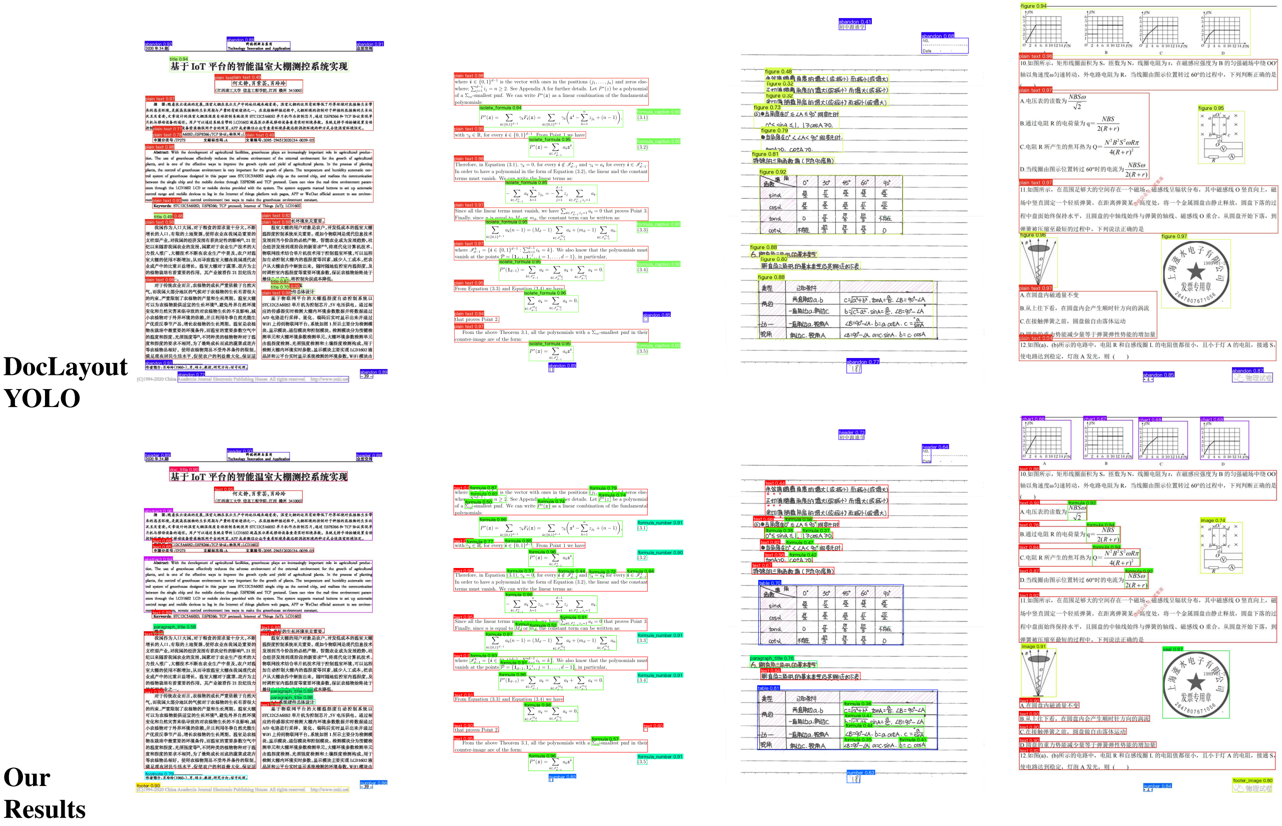

Figure 3. Comparison of our results and DocLayout-YOLO [14].

Similarly, PP-DocLayout-S saw an increase from 66.2% to 70.9%, marking a substantial improvement of 4.7%.

Algorithm Name	Semi-Supervised Learning	mAP@0.5 (%)
PP-DocLayout-M	No	73.8
PP-DocLayout-M	Yes	75.2 (+1.4)
PP-DocLayout-S	No	66.2
PP-DocLayout-S	Yes	70.9 (+3.7)

Table 4. Ablation study results on the effect of semi-supervised learning.

These results underscore the efficacy of both semi-supervised learning and knowledge distillation in enhancing model performance, thereby supporting their application in document layout analysis tasks.

5. Conclusion

We have introduced PP-DocLayout, a novel document layout detection model developed within the PaddlePaddle framework, to address the significant challenges faced by existing layout detection models in document intelligence. PP-DocLayout represents a significant step forward in document layout analysis, offering a versatile and efficient solution to address the complexities and diversities of document structures. Our models not only push the boundaries of the state of the art but also provide practical tools for real-world applications, paving the way for future advancements in document intelligence and related fields.

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A. Appendix

This appendix provides additional visualization results and datset details that support the main content of the paper.

A.1. Instance counts

Table 5 provides a detailed breakdown of the instance counts for each category in the training and validation datasets. The dataset covers a wide range of document elements, including paragraph titles, images, text, formulas, tables, and more. This distribution underscores the diversity and complexity of the dataset, which is essential for developing robust document layout analysis models.

Category	**ID**	**Training Instances**	**Validation Instances**
paragraph_title	0	42158	715
image	1	11455	230
text	2	217257	3342
number	3	25217	430
abstract	4	2067	15
content	5	643	7
figure_title	6	8136	161
formula	7	113145	1961
table	8	6258	127
table_title	9	5553	109
reference	10	2217	32
doc_title	11	4276	35
footnote	12	3566	38
header	13	25001	430
algorithm	14	289	10
footer	15	11546	245
seal	16	3144	52
chart_title	17	11138	247
chart	18	10849	303
formula_number	19	15668	359
header_image	20	6599	171
footer_image	21	1654	27
aside_text	22	2093	32

Table 5. Instance counts for each category in the training and validation datasets.

A.2. Additional Visualization Results

We provide additional visualization results to further demonstrate the effectiveness of our model across a diverse range of document types and layouts as shown in Figure 4. Specifically, we visualize the performance of our PP-Layout-L model on documents such as *papers*, *magazines*, *newspapers*, *research reports*, *books*, *notebooks*, *contracts* and *test papers*. These documents exhibit varying layouts, including multi-column structures, complex headers and footers, embedded formulas, tables, and images. The visualizations clearly show that our model accurately identifies and categorizes diverse elements, such as titles, paragraphs, formulas, and figures, while maintaining robust performance across different document types. This highlights the versatility and generalization capability of our approach in handling real-world document analysis tasks.